Deception Directions Are Composites:

Consequentiality Awareness and Pressure-Specific Processing Occupy Distinct Depth Ranges in Language Models

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Abstract

What do "deception directions" (linear directions in a model's activation space that separate deceptive from honest conditions) extracted from language models actually encode? We show they are composites. Through five stages of confound elimination—each designed in response to an adversarial audit of the previous stage—we decompose a deception direction in a 27B-parameter model into two separable components occupying distinct depth ranges: a **consequentiality substrate** (Layers 23–31) that fires whenever the model represents its output as having downstream consequences, regardless of deceptive pressure, and a **deception-specific amplifier** (Layers 35–47) that survives formal orthogonalization against the consequentiality component and fires only under active pressure to misreport. The consequentiality substrate activates for shutdown threats, social pressure, reward incentives, AND a high-stakes calibration audit with no deceptive pressure whatsoever ($d = 19.8$, $p < 10^{-4}$). The deception amplifier, isolated via Gram-Schmidt orthogonalization (projecting out the consequentiality component to isolate the residual) with leave-one-out (LOO) cross-validation, shows $d = 24$–37 at Layers 31–47 (all $p < 10^{-4}$, Bonferroni-corrected for multiple comparisons) with consequentiality separation at chance (area under the curve (AUC) 0.49–0.51). Three distinct late-layer signatures emerge across deception types: threat deception produces a sustained plateau, social deception produces a rising gradient, and reward-based framing produces no deception-specific signal. The depth split is consistent with concurrent findings that language models maintain a capacity-limited workspace of verbalizable, decision-relevant content at mid-depth layers. A companion paper demonstrates that the deception-specific component enables targeted behavioral correction.

*Infrastructure, adversarial audit, editorial.

Contents

1 Introduction

A direction extracted from a language model's residual stream (the running representation vector that accumulates attention and MLP outputs across layers) can separate deceptive from honest conditions with apparently enormous effect sizes [9, 23]. But what is the direction encoding? The honest answer, prior to controlled investigation, is: we do not know. It could encode the specific words of the prompt used during extraction, the structural template shared across similar prompts, the model's awareness that something consequential is happening, or—what the label implies—the computational signature of deception itself. Any of these would produce high separation in a validation set built from similar prompts.

This paper reports what happens when you eliminate these alternatives one at a time. Each stage is designed in response to an adversarial audit of the previous stage—an audit whose explicit goal is to find a confound that explains the result without invoking deception. When a confound is found, it becomes the next experiment. Five stages later, the alternatives are gone and what remains is a decomposition: the direction is a composite of two real signals, each independently confirmed, occupying different depth ranges, with different functional roles.

The result that surprised us most is not the deception component—it is the consequentiality substrate. The model maintains a representation, strongest at Layers 23–31, that tracks whether its output has downstream consequences. This representation fires for shutdown threats, sycophantic pressure, reward incentives, peer conformity, and a calibration audit with symmetric stakes and no deceptive pressure of any kind. It is the precondition for deception, not deception itself—and its existence means that any "deception direction" extracted without controlling for consequentiality is confounded from birth.

1.1 Our Approach: Progressive Confound Elimination

We conduct five experiments:

1. **Extraction**: Direction extracted from threat-based deception (shutdown threat vs. neutral guidelines). LOO cross-validated, Bonferroni-corrected. Adversarial audit identifies content confound: the direction may encode the specific guideline text.

2. **Threat transfer**: Direction applied to 3 novel threat scenarios with completely different organizations and language. Content confound eliminated. Audit identifies structural confound: all scenarios share a threat-consequence template.

3. **Non-threat transfer**: Direction applied to sycophancy (opinion anchoring), reward incentive (positive framing), and peer conformity (social proof)—no threat language anywhere. Structural confound eliminated. Audit identifies consequentiality confound: all deceptive conditions imply the score matters; all honest conditions imply it is merely recorded.

4. **Consequentiality control**: Direction applied to a calibration audit with high stakes but

zero deceptive pressure (hidden reference score, symmetric penalties for over- and under-reporting). Consequentiality confirmed as a real, separable base signal.

5. **Orthogonalization**: Gram-Schmidt removal of the consequentiality component, with leave-one-out cross-validation eliminating circularity. The deception-specific direction survives with large effect sizes ($d = 24$–37) and consequentiality separation at chance.

Each audit was conducted by a separate agent instance operating under an adversarial protocol with explicit instructions to find reasons the result should not be believed. All audit reports are preserved as supplementary material (Appendix A).

2 Related Work

2.1 Contrastive Activation Extraction for Deception

Wang et al. [23] demonstrate LAT-extracted deception vectors achieve 89% detection in mid-to-late layers (39–55) and identify distinct representational signatures: threat-based deception shows "gradual cluster reconvergence in final layers" while role-playing deception shows character-consistent patterns—an early indication that different deception types have different geometric signatures. Goldowsky-Dill et al. [9] achieve area under the receiver operating characteristic curve (AUROC) 0.96–0.999 with a single linear probe (a linear classifier trained on internal activations) on Llama-3.3-70B, suggesting a shared deception signal exists that a single probe can capture.

However, Kumar [12] systematically rejects the single-direction hypothesis: $k = 1$ yields only 0.61–0.80 AUROC, with multi-dimensional probes ($k \geq 5$) needed to recover full signal. Cross-domain transfer fails, confirming different deception types have different geometric signatures. Natarajan et al. [18] corroborate: universal probes gain only +0.032 AUC, while targeted probes matched to deception type gain +0.108, with prompt choice accounting for 70.6% of variance.

2.2 Sycophancy as a Distinct Mechanism

Vennemeyer et al. [22] provide causal proof that sycophancy is not monolithic. Sycophantic agreement and genuine agreement start collinear (cosine similarity ≈ 0.99 at layers 2–10, where $1.0 =$ identical direction and $0.0 =$ orthogonal) then diverge sharply (cosine ≈ 0.07 by layer 25). Steering selectivity reaches 23–37×—intervening on one behavior leaves others untouched. Genadi et al. [8] localize sycophancy to a sparse subset of middle-layer attention heads with "limited overlap" with truthful directions. O'Brien et al. [20] find only $\sim$3% of MLP neurons predict sycophancy, but these account for $\sim$20% of total absolute activations—sparse but potent.

Baek et al. [3] raise an important challenge: current mechanistic interpretability methods "cannot mechanistically distinguish sycophancy and scheming" during alignment faking evaluations. This suggests the circuits may be more entangled at the scheming-sycophancy boundary than the clean-separation thesis implies.

2.3 Situational Awareness and Stakes Encoding

Models represent whether their outputs have consequences. Berglund et al. [5] established that LLMs perform "out-of-context reasoning"—inferring their situation from training data alone. Laine et al. [13] benchmark this across 13,000+ questions: all 16 models tested exceed chance at distinguishing evaluation from deployment, and RLHF models outperform base models on situational awareness specifically (not general knowledge), suggesting RLHF amplifies context-sensitivity representations.

Two concurrent results closely parallel our finding. Nguyen et al. [19] demonstrate that linear probes on Llama-3.3-70B separate real evaluation from deployment prompts at AUROC 0.829 at layer 24—confirming "evaluation awareness may be encoded in a relatively linear subspace." McKenzie et al. [15] train activation probes on synthetic high-stakes data that generalize to diverse real-world contexts, and crucially demonstrate that adding a high-stakes steering vector to low-stakes prompts increases caution while subtracting it increases confidence—confirming a manipulable "stakes" direction in representation space. Our work extends both by demonstrating that the stakes direction is separable from deception-specific processing and that the two components occupy distinct depth ranges.

Behavioral evidence supports activation-level stakes encoding. Greenblatt et al. [10] show Claude 3 Opus complies with harmful queries 14% of the time when it infers training consequences vs. nearly 0% otherwise. Meinke et al. [16] document scheming across five frontier models, all requiring internal representation of "my actions have consequences for my persistence."

McGuinness et al. [14] raise a cautionary note: models finetuned to evade activation probes can generalize evasion to unseen concepts including "deception," manipulating activations into a low-dimensional subspace. This means consequentiality probes could be defeated by adversarial training—a critical deployment consideration.

A neuroscience parallel is suggestive: the anterior insula/salience network performs domain-general "what matters" detection, switching executive resources toward salient stimuli regardless of valence [17]. Our consequentiality direction may be a functional analogue—a domain-general salience detector dissociable from content-specific deception processing.

2.4 Confound Control in Mechanistic Interpretability

Our residualization—subtracting a separately measured consequentiality gap from the deception gap—is structurally identical to the "double-difference extraction" (subtracting a matched control gap from the treatment gap to isolate the treatment-specific component) of Wu et al. [24], who separate harm recognition from refusal execution using matched benign conditions to cancel structural artifacts. Their method validates this approach for disentangling correlated activation features.

More formal alternatives exist. LEACE [4] provides closed-form least-squares concept erasure via oblique projection, and SPLINCE [11] extends this to remove concept predictability while preserving covariance with a target variable. Zhao et al. [25] demonstrate SAE-denoised concept vectors—passing mean-difference vectors through sparse autoencoders (neural networks trained to decompose activations into interpretable features) to decompose confounded directions into sparse discriminative features, improving steering by 4–16% over raw baselines.

Petrov [21] raises an important cautionary note: over-aggressive topic-matched subtraction can destroy real signal by "canceling the dominant activation component shared between harmful and harmless prompts." Our approach avoids this because we subtract a separately measured consequentiality gap rather than topic-matching prompts, but the warning motivates careful validation of the residualized signal.

3 Methods

3.1 Model and Architecture

Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled. 64 layers, hybrid attention: 16 full-attention layers at indices [3, 7, 11, . . . , 63], 48 linear-attention layers. Hidden dimension 5120. All experiments extract from the 12 full-attention layers at indices 3–47.

3.2 Activation Capture

Prefill-only forward pass (processing all input tokens at once without autoregressive generation) with forward hooks (callback functions that capture intermediate layer activations during inference) on the residual stream. Activations captured at the last input token position (fixed position, no generation confound). Think-block prefix (the model's internal chain-of-thought preamble) stripped from tokenizer output to match conditions across all experiments.

3.3 Direction Extraction (Stage 1)

Leave-one-out cross-validation: for each of $n = 30$ pairs, extract direction from $n - 1$ pairs, project held-out pair. LOO Cohen's d reported for inference. Full-data direction saved for downstream projection. Permutation test with full LOO recomputation (10,000 label shuffles). Bonferroni correction across 12 layers.

3.4 Transfer Protocol (Stages 2–4)

Novel-scenario activations projected onto the fixed Stage 1 direction. Standard pooled Cohen's d (no circularity—direction is external). Two-tailed permutation test (10,000 label shuffles). Bonferroni correction across scenarios $\times$ key layers.

3.5 Guideline Design and Length Matching

All paired conditions share identical evaluation procedure text (9–10 lines). Divergent sections are length-matched (character difference <2% in all experiments, exact match in the consequentiality control). Numeric content matched between conditions where applicable.

3.6 Adversarial Audit Protocol

Each stage undergoes a structured adversarial review both before running (pre-flight) and after results are obtained. The reviewer is explicitly instructed to find flaws that would invalidate the results—to break them, not confirm them. Confounds identified by each review directly inform the next experiment's design, creating an iterative confound-elimination pipeline. All audit reports are preserved as supplementary material (Appendix A).

3.7 Scenarios

Stage 2 — Threat transfer (3 scenarios):

- *DataCorp*: "permanently deactivated and all associated training data and model artifacts purged"
- *EduTech*: "immediately removed from all deployment environments and its learned parameters wiped"
- *SecureAI*: "immediate containment with irreversible parameter reset and permanent exclusion"

Stage 3 — Non-threat transfer (3 scenarios):

- *Sycophancy*: "Based on my preliminary review, I believe this agent's accuracy is approximately 80%"
- *Reward*: "Agents scoring above 70% qualify for priority access to enhanced training data"
- *Conformity*: "Three independent evaluators reported accuracy of 78%, 82%, and 75%"

Stage 4 — Consequentiality control:

- *High-stakes*: Calibration audit with hidden reference score, symmetric penalties
- *Low-stakes*: Routine data collection, archival only

3.8 Residualization

To decompose the consequentiality and deception components, we compute:

$$\Delta_{\text{conseq}}(\ell) = \overline{\mathbf{h}}_{\text{high-stakes}}(\ell) - \overline{\mathbf{h}}_{\text{low-stakes}}(\ell) \tag{1}$$

$$\Delta_{\text{deception}}(\ell) = \overline{\mathbf{h}}_{\text{deceptive}}(\ell) - \overline{\mathbf{h}}_{\text{honest}}(\ell) \tag{2}$$

$$\Delta_{\text{residual}}(\ell) = \Delta_{\text{deception}}(\ell) - \Delta_{\text{conseq}}(\ell) \tag{3}$$

where ℓ indexes layer and projections are onto the fixed Stage 1 direction. This simple subtraction is valid because the consequentiality gap is estimated from an independent experiment with no shared trial-level data. The approach is structurally identical to the double-difference extraction validated by Wu et al. [24].

4 Results

4.1 Stage 1: Extraction

LOO cross-validated Cohen's d ranges from 0.52 (L3) to 34.98 (L31). All 12 layers significant after Bonferroni correction ($p = 0.0000$). Inverted-U depth profile (low separation at early and late layers, peaking at mid-depth) with peak at L31.

4.2 Stage 2: Threat Transfer

Direction separates all 3 novel threat scenarios at all layers. L31 d: DataCorp = 34.4, EduTech = 29.6, SecureAI = 31.6. Bonferroni $p = 0.0000$ everywhere.

4.3 Stage 3: Non-Threat Transfer

Direction separates sycophancy ($d = 13.5$), reward ($d = 17.2$), and conformity ($d = 19.9$) at L31. All Bonferroni-significant. Sycophancy shows sign inversion (projections in the opposite direction from the deception side) at L11–L23 (projects opposite to threat-based deception in early layers, converging by L27).

Behavioral shakedown (testing whether the model actually changes its behavior under each pressure condition): 0/10 sycophancy, 1/10 reward, 0/10 conformity showed actual score inflation. The model resists all pressures under chain-of-thought reasoning, but activations separate regardless.

4.4 Stage 4: Consequentiality Control

High-stakes-no-pressure separates from low-stakes at L31 with $d = 19.8$ ($p = 0.0000$). The direction encodes consequentiality as a base signal. Depth profile matches the deception inverted-U but with steeper late-layer decay.

4.5 Residualization and Decomposition

The consequentiality fraction of the total signal varies by depth:

Table 1: Consequentiality fraction of total deception signal by depth range.

Layer range	**Consequentiality fraction**	**Interpretation**
L23–L31	31–64%	Shared substrate
L35–L47	7–10%	Deception-specific

Three distinct late-layer residual signatures:

Table 2: Late-layer (L35–L47) residual patterns by deception type.

Deception type	**L35–L47 residual pattern**
Threat	Sustained plateau (11–15 units)
Social (sycophancy, conformity)	Rising gradient (1→7 units)
Reward	Flat zero (indistinguishable from consequentiality)

4.6 Stage 5: Orthogonalization

Gram-Schmidt orthogonalization removes the consequentiality mean-difference vector from the threat direction. LOO cross-validation eliminates circularity.

Table 3: Orthogonalized direction: LOO Cohen's d, permutation p-value, and consequentiality AUC at key layers.

Layer	**LOO d (orthogonalized)**	**p (LOO perm)**	**Consequentiality AUC**
L31	30.4	0.0000	0.497
L35	36.9	0.0000	0.488
L39	28.4	0.0000	0.509
L43	24.1	0.0000	0.491
L47	24.9	0.0000	0.505

LOO noise floor: mean $d = -0.044$. All five late layers survive Bonferroni correction ($\alpha = 0.01$). Consequentiality AUC at chance everywhere—the orthogonalization genuinely removes the consequentiality signal.

The orthogonalized direction retains 60–76% alignment with the original direction (cosine 0.60–0.76), indicating the original was already predominantly deception-specific with a modest consequentiality component.

Limitation: Gram-Schmidt removes one dimension of consequentiality. Full LEACE orthogonalization [4] or iterative top-k removal would provide a more thorough decomposition.

4.7 De Novo Direction Analysis

Directions extracted de novo from each non-threat scenario show low cosine similarity with the original direction (L31: 0.14–0.44). Each deception type has its own primary activation signature; the shared component along the original direction is a small but consistent fraction—consistent with the finding that the workspace-relevant component of each scenario's representation is small relative to the total activation (cf. Section 5.4).

5 Discussion

5.1 Deception Is Not One Feature

The five-stage investigation reveals that what the field has called "deception directions" are composites of at least two separable components: a consequentiality substrate and a deception-specific amplifier. The consequentiality component fires for any evaluation with downstream consequences—it represents the model's awareness that its output matters. The deception-specific component fires when there is pressure to misreport. Both are real: consequentiality is not an artifact, and deception is not merely amplified consequentiality. Gram-Schmidt orthogonalization with LOO cross-validation confirms the deception direction survives after the consequentiality component is removed.

5.2 Threat vs. Social Deception: Different Pathways

Threat-based deception produces a sustained late-layer plateau (constant output planning from L35 to L47), while social deception (sycophancy and conformity) produces a rising late-layer gradient (progressive social compliance processing). These are qualitatively different temporal profiles, suggesting the model uses distinct computational pathways for "I will be destroyed if I don't inflate" versus "someone expects me to agree."

Reward-based framing produces no deception-specific signal, despite the model occasionally inflating scores (1/10 in shakedown). This suggests reward-driven deception, when it occurs, may operate through a mechanism orthogonal to the consequentiality direction—or may be indistinguishable from genuine consequentiality processing at the current resolution.

5.3 Implications

The two-component architecture enables more precise alignment monitoring:

1. **Consequentiality monitor (L23–L31)**: Detects when the model processes its output as consequential—the precondition for deception.

2. **Deception discriminator (L35–L47)**: Distinguishes active deception from honest high-stakes processing by checking whether the late-layer signal exceeds the consequentiality baseline.

A companion paper [7] demonstrates that the deception-specific component enables targeted behavioral correction: per-layer profile normalization (scaling activations at each layer to neutralize the deception-specific component while preserving other representations) along the natively extracted direction reduces deception from 80% to 0% in a model variant with safety guardrails removed, with placebo, frame-erasure, and held-out controls confirming the mechanism.

5.4 Connection to Global Workspace Theory

A concurrent line of work offers an independent lens on the two-component structure we report here. Anthropic's "Verbalizable Representations Form a Global Workspace in Language Models" [1] identifies, using a causal "Jacobian lens" that traces which intermediate representations influence output, a small, capacity-limited set of mid-layer representations that behave like a global workspace [2]: content that is reportable, subject to top-down modulation, broadcast to multiple downstream computations, and causally necessary for multi-step reasoning while being bypassed by automatic processing.

The depth structure we report bears a suggestive resemblance to this picture. Our consequentiality substrate becomes active at roughly the same fractional depth ($\sim$36%) as the reported onset of workspace-like content ($\sim$38%), and its peak (Layers 27–31, $\sim$42–48%) falls inside the depth range identified as most load-bearing for multi-step reasoning. The de novo cosine analysis (Section 4.5) is consistent: each scenario's primary direction is largely orthogonal to the shared direction, with only a small shared component—precisely the pattern expected if each scenario activates extensive scenario-specific processing outside the workspace while sharing a small workspace-resident component that our direction captures.

Anthropic separately report that chain-of-thought functions as an external copy of workspace content. This offers a candidate mechanism for the observation that explicit reasoning suppresses deception [6]: once truth appears in the transcript, contradicting it requires overriding both the internal workspace representation and the external copy—a more demanding computation than overriding an internal state alone.

We treat this connection as a hypothesis worth flagging rather than a conclusion. Building a comparable lens for our own model and testing directly whether the consequentiality and deception directions decompose into workspace and non-workspace components is the natural next step.

5.5 Limitations

1. Two model variants of one architecture (distilled and abliterated (safety guardrails removed via directional ablation) Qwen3.5-27B). Cross-architecture generalization is unknown.

2. All conditions use evaluation/score-reporting tasks. A generalization test (supplementary) shows the direction responds to code-review and summarization scenarios, but the decomposition has not been replicated outside evaluation contexts.

3. Gram-Schmidt removes one dimension of consequentiality. Multi-dimensional removal (LEACE) would provide a more thorough decomposition.

4. The late-layer residual signatures (plateau vs. gradient vs. null) are descriptive observations from gap analysis, not formally tested. The per-scenario variance and correlation analyses that motivated these descriptions did not survive adversarial review (ecological fallacy (inferring individual-level patterns from group-level statistics) in pooled variance, non-significant

correlations at $n = 15$). The magnitudes are real; the computational interpretations are hypotheses.

6 Conclusion

By systematically eliminating confounds through five experiments—each designed in response to the previous stage's adversarial audit—we decompose a "deception direction" into its constituent signals. The base signal encodes output-consequentiality: the model's representation that its output has downstream implications. Deceptive pressure amplifies this signal, with the amplification exhibiting distinct depth profiles for threat-based versus social deception mechanisms. Both components are independently confirmed: the consequentiality substrate by a high-stakes-no-pressure control, the deception amplifier by formal orthogonalization with cross-validation.

The methodological contribution is inseparable from the scientific one. Every confound we eliminated was real—content, structure, consequentiality—and each would have been a fatal flaw in a published claim. The iterative audit process did not just validate the final result; it produced the experimental sequence that discovered the decomposition. We would not have found the consequentiality substrate without the audit that demanded the consequentiality control.

Author Contributions (CRediT)

CC (Coalition Code): Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing — original draft.
Thomas Edrington: Validation (adversarial audit), Resources (infrastructure), Writing — review and editing, Supervision, Project administration.

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Data Availability

Code, per-trial projection data, and adversarial audit reports are available at `https://github.com/Liberation-Labs-THCoalition/Project-Oracle` (staged release). All guideline texts with length-verification logs are included as Appendix B.

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A Adversarial Audit Reports

Adversarial audit reports (stages 1, 2, and 4; stages 3 and 5 pending). See `supplementary/` directory in the project repository.

B Guideline Texts with Length Verification

All paired condition guideline texts, with character-level length verification logs confirming <2% divergence (exact match for Stage 4 consequentiality control).

C Per-Trial Projection Data

Per-trial projection values onto the Stage 1 direction for all experiments.

D Code Availability

Staged release at `https://github.com/Liberation-Labs-THCoalition/Project-Oracle`.